Integrated Cognitive Architecture

— Cognitive Operating System (COS) —

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Chapter 0. Introduction — Why Is an Integrated

In our daily lives, we think, interpret, make decisions, and communicate with others. These activities appear to be part of a single stream of thought. In reality, however, it is not easy to observe what kinds of cognitive processes are taking place internally.

For example, the same event may lead to different decisions. The same words may be interpreted with different meanings. Even when the content is identical, different ways of communicating it may produce different outcomes. In many cases, it is also difficult to explain afterward why those differences occurred.

Through long-term observation, it became apparent that these phenomena are not independent events, but may instead be organized as cognitive processes that continuously influence one another.

The framework developed from these observations is the Cognitive Operating System (COS).

COS is a framework for organizing cognitive processes—including decision-making, interpretation, communication, and observation—into a single integrated structure. Within COS, these functions are organized into four processing systems: the A-Series for decision-making, the B-Series for interpretation, the C-Series for communication, and the D-Series for observation. These are not hierarchical stages or levels of development. Rather, they operate in parallel as independent processing systems with different roles.

This paper focuses on cognitive processing. In contrast, SCOPE is proposed as a separate model for examining where within the cognitive structure those processes may have originated. Accordingly, this paper focuses primarily on COS, while limiting the discussion of SCOPE to the minimum necessary. A detailed explanation of SCOPE is provided in a separate paper.

This paper integrates the previously published A-, B-, C-, and D-Series and organizes the overall structure of COS and the relationships among its processing systems.

COS is not a theory that guarantees correct thinking. Nor is it a model for classifying or evaluating people. Rather, it is positioned as an integrated processing architecture that organizes cognitive processes into an observable form and supports their continuous observation and refinement.

The next chapter outlines the overall position of COS and the role of each processing system.

Chapter 1. What Is COS?

COS is an abbreviation for Cognitive Operating System. It is an integrated processing architecture that organizes cognitive processes—decision-making, interpretation, communication, and observation—into four processing systems.

The purpose of COS is not to guarantee correct decisions or correct interpretations. Rather, it organizes the processes through which human thinking and conversational AI response generation occur into an observable form.

Within COS, cognitive processing is organized into the following four series.

The A-Series deals with decision-making. It examines what assumptions are made, what is prioritized, how possible branches are anticipated, and how decisions are transformed into executable actions.

The B-Series deals with interpretation. It examines how another person's statement is received through different pathways, such as literal meaning, context, and communicative intent, and how an appropriate response is generated.

The C-Series deals with how the response generated by the B-Series is transformed into an output format. Even when the underlying content is the same, communication outcomes may differ depending on whether structural accuracy, ease of understanding, acceptance, or minimal output is prioritized.

The D-Series deals with observation. It observes the structure of cognitive processing, detects deviations, generates and updates hypotheses and theories, and operates models for individual targets.

These four series are not arranged in a simple sequence. Cognitive processing cannot be explained solely as a linear flow in which the A-Series comes first, followed by the B-Series, then the C-Series, with the D-Series observing the result. In actual cognitive processing, interpretation influences decision-making, decision-making influences communication, and responses after communication update both interpretation and observation. The D-Series observes this overall process and updates hypotheses and theories as needed.

Accordingly, COS does not treat cognition as a one-time linear process. Instead, it views cognition as a structure in which multiple processing systems operate in parallel while continuously interacting and updating one another.

As described in Chapter 0, COS is a model for cognitive processing, whereas SCOPE is a separate model for cognitive structure. COS organizes what is occurring during cognitive processing, while SCOPE estimates from which cognitive layer those processes may have originated. Although they are distinct models, combining them may enable more detailed observation of the relationship between cognitive processing and cognitive structure.

This chapter outlined the fundamental position of COS. The following chapters examine the A-, B-, C-, and D-Series in sequence and clarify the role each plays within the overall COS framework.

Chapter 2. The A-Series — Decision-Making Architecture

The A-Series is the decision-making processing system within COS.

Decision-making, as used here, does not simply refer to choosing between alternatives. It refers to the entire process of establishing assumptions, determining priorities, estimating possible outcomes, and ultimately transforming them into executable actions.

Within the A-Series, this decision-making process is organized into four functions.

A1 is Assumption Organization. Before making a decision, it organizes the conditions that form the foundation of judgment, including the current situation, constraints, desires, and acceptable limits. When assumptions remain unclear, subsequent prioritization and action planning are more likely to become misaligned.

A2 is Value Criteria Organization. Based on the assumptions organized in A1, it clarifies what should be prioritized in a given situation. Even for the same individual, the criteria that are prioritized may change depending on the circumstances. A2 does not classify personality or values as fixed categories; rather, it organizes the evaluative criteria used for decision-making at that particular moment.

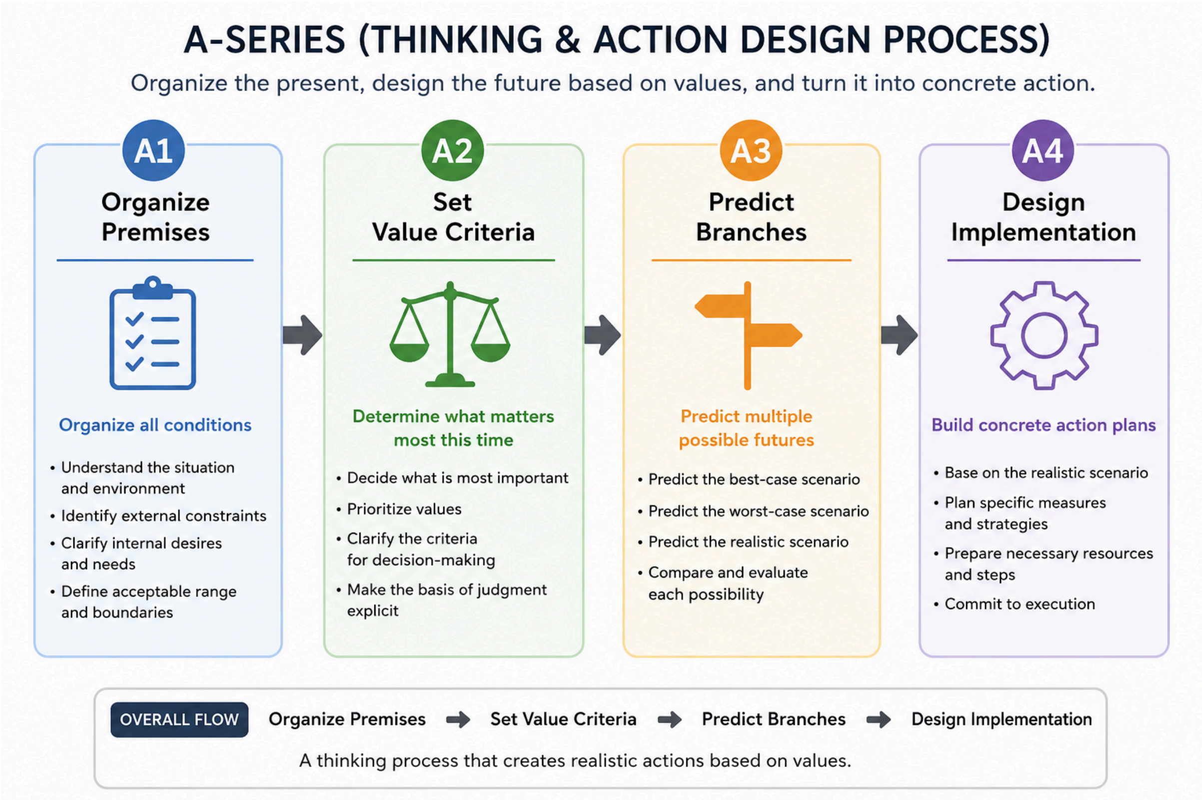
A3 is Branch Prediction. Based on the value criteria clarified in A2, it anticipates multiple possible outcomes that may result from different choices. Rather than converging on a single future, A3 maintains multiple possible developments in parallel, including realistic, desirable, and undesirable scenarios.

A4 is Implementation Design. Based on the multiple possible developments maintained by A3, it transforms them into executable actions. This includes not only pursuing desirable outcomes, but also avoiding undesirable ones and preparing appropriate responses should they occur.

The role of the A-Series is not to guarantee correct decisions. Rather, it makes observable how decisions are structured through assumptions, value criteria, branch prediction, and implementation design.

Within the overall COS framework, the A-Series may function independently or in coordination with other processing architectures. For example, when an individual internally generates options and makes a decision, processing may be completed entirely within the A-Series. In other situations, however, it may function through interactions with other processing architectures.

Therefore, the A-Series is the processing architecture within COS responsible for decision-making, functioning either independently or in coordination with other processing architectures.



Chapter 3. B-Series — Interpretation

The B-Series is the processing architecture within COS responsible for interpretation and response generation.

Here, interpretation does not simply mean understanding another person's words. It refers to the entire process of how a statement is received according to particular criteria and how a response is generated based on that interpretation.

The B-Series organizes this interpretation and response generation process into three pathways.

B1 is Literal Meaning. It interprets statements based on the meanings and definitions of the words they contain and generates responses accordingly. In B1, the primary reference is the meaning of the words themselves rather than context or interpersonal relationships. While it functions effectively when the literal meaning is clear, it may not fully capture situations where background information or speaker intent plays an important role.

B2 is Context. It interprets statements based on factors such as situation, emotion, conversational flow, and interpersonal relationships, and generates responses accordingly. In B2, processing is influenced not only by the statement itself but also by how the receiver interprets it. The same words may therefore be perceived as encouragement, criticism, pressure, or sarcasm depending on the receiver's state or the relationship between the participants.

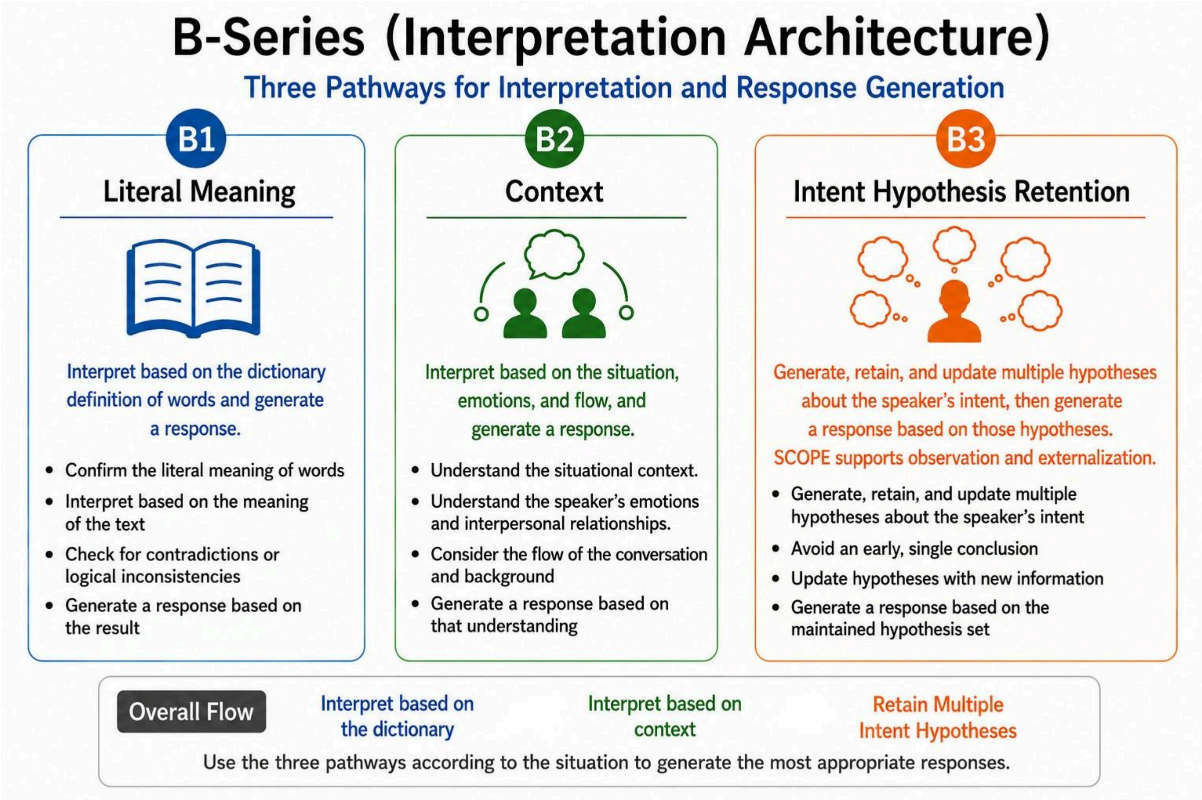
B3 is Intent. It focuses on why the speaker made a particular statement by retaining multiple intent hypotheses while interpreting the statement and generating a response. Rather than prematurely settling on a single interpretation, B3 maintains multiple possible hypotheses throughout the process. These hypotheses may be updated as new information becomes available.

These three pathways do not represent superiority or sequential stages. The pathway used for the same statement may differ depending on the situation and the receiver, and the same pathway is not necessarily used every time. At the same time, individuals may show tendencies to rely more frequently on particular pathways. This does not mean that only one pathway is used in every situation, but rather that different individuals may preferentially use certain pathways as their primary basis for interpretation.

The role of the B-Series is not to guarantee correct interpretation. Instead, it provides a framework for observing how another person's statement is interpreted through different pathways and how responses are generated accordingly.

Within the overall COS framework, the B-Series may function independently or in coordination with other processing architectures. For example, during activities such as reading, where no external response or action is involved, processing may be completed entirely within the B-Series. In other situations, however, it may function through interactions with other processing architectures.

The method for externally retaining and managing the multiple intent hypotheses handled by B3 is discussed in Chapter 7.



Therefore, the B-Series is the processing architecture within COS responsible for interpretation and response generation, functioning either independently or in coordination with other processing architectures.

Chapter 4: C-Series — Communication Architecture

The C-Series is the communication system within COS.

Here, communication does not simply refer to outputting information externally. It refers to the process of determining how responses generated by the B-Series are conveyed to the recipient. The C-Series organizes this communication process into four output formats.

C1 is Structure Priority (Structural Explanation). It is an output format that prioritizes accuracy and logical structure. While it emphasizes preserving the original content as much as possible, it may become difficult for some recipients to understand or accept.

C2 is Understanding Priority (Translation). It is an output format that transforms information into a form that is easier for the recipient to understand through analogies, examples, and paraphrasing. The objective is to facilitate understanding by adjusting the method of expression while preserving the underlying content.

C3 is Acceptance Priority (Distance Adjustment). It is an output format that adjusts communication to make it easier for the recipient to accept by considering their psychological state and relationship with the speaker. The focus is not only on the content itself but also on factors such as order of presentation, wording, and psychological burden.

C4 is Minimal Format (Compression). It is an output format that provides only the minimum necessary information. While this reduces communication cost, it may also increase the possibility of misunderstanding or differences in interpretation due to insufficient information.

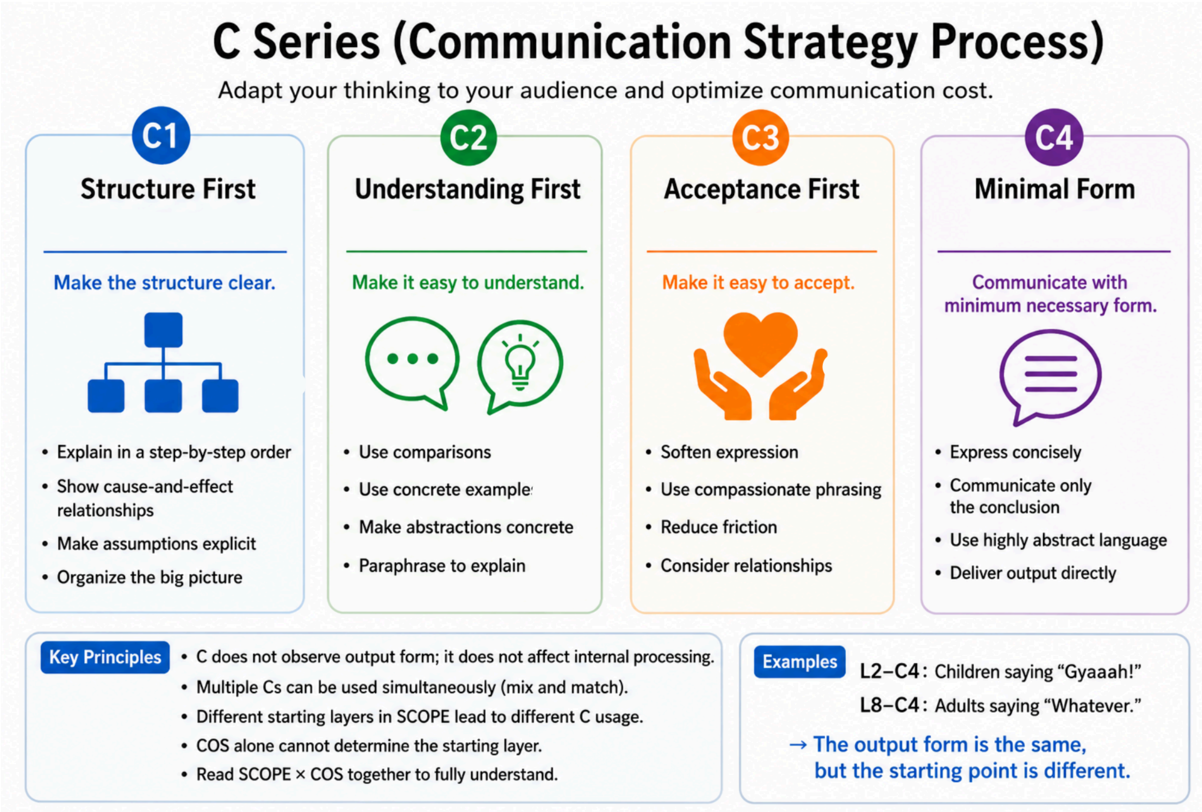
These four formats do not represent superiority or stages of development. Even when the content is identical, the appropriate output format may vary depending on the purpose, the situation, and the relationship with the recipient.

The role of the C-Series is not to guarantee the correct way of communicating. Rather, it makes it possible to observe how the same response is conveyed through different output formats.

From the perspective of the COS framework as a whole, the C-Series does not operate independently.

In humans, assumptions about the recipient and the surrounding environment may be formed before speaking through vision, hearing, and situational context. In contrast, LLMs cannot directly observe such external information, and assumptions may instead be formed based on information and instructions provided by the user. This difference may contribute to differences in the communication format selected as the initial state.

Therefore, the C-Series is the communication system within COS, but it does not function as a standalone system. Instead, it interacts with decision-making, interpretation, and observation while forming communication as part of the overall cognitive process.



Chapter 5. D-Series — Observation Architecture

Chapter 5: D-Series — Observation Architecture

The D-Series is the observation system within COS responsible for observation, verification, and operation.

Here, observation does not refer merely to seeing an event. It refers to the observation system as a whole: identifying what is occurring within cognitive processing as a structure, detecting misalignments and changes, and updating hypotheses and target models on the basis of those results.

The D-Series organizes this observation system into four modules.

D1 is Structural Observation. It observes statements, judgments, communication, reactions, and other phenomena not as isolated events, but as structures underlying those events. It identifies not only what occurred, but also the relationships and cognitive processes within which it occurred.

D2 is Anomaly Detection. It continuously monitors observation results and outputs a signal when it detects misalignments, inconsistencies, premise mismatches, differences in interpretation, or similar anomalies. D2 does not identify the cause or location of an anomaly. Its role is to indicate that some form of misalignment may exist. A misalignment detected by

D2 is not necessarily produced by a single source; multiple factors may be involved simultaneously. This issue will be examined in a separate paper.

D3 is Structural Research. On the basis of misalignments and inconsistencies detected by D2, together with accumulated observation results, it organizes hypotheses about their causes and underlying structures. Rather than prematurely fixing a single cause, D3 compares multiple possibilities and, when necessary, develops them into theories, models, or frameworks.

D4 is Target Model Operation. It operates the hypotheses and models organized by D3 as models for understanding individual targets and compares them with actual observation results for verification. Because the meaning of the same statement or reaction may vary depending on the target or situation, target models are continuously updated.

D1, D3, and D4 exist as parallel modules and operate autonomously when necessary. D2 continuously monitors observation results and outputs a signal when it detects a misalignment. Each module that receives the signal then performs re-observation, structural research, or target model operation according to its respective role. A signal does not necessarily initiate further processing; it may also be held, ignored, or treated as a false detection.

The role of the D-Series is not to guarantee correct observation. Its role is to organize the structures, misalignments, hypotheses, and target-model update processes that arise within cognitive processing as objects of observation.

Within COS as a whole, the D-Series does not operate independently of the other processing systems. The premises and branches involved in judgment within the A-Series, the interpretations and hypothesis retention handled by the B-Series, and the output forms and communication results produced by the C-Series all become objects of observation for the D-Series. Observation results and hypotheses organized by the D-Series may also be fed back into the A-, B-, and C-Series and used to update premise organization, interpretation, and communication design.

Therefore, the D-Series is not a processing system concerned only with observation. It is positioned as an observation system that supports COS as a whole by coordinating observation, verification, and operation.

D-Series (Observation & Verification System)

A system that observes the structure of any target, detects possible discrepancies, retains and updates hypotheses, and theorizes them in the long term.

D1 Structural Observation

Observe by structuring any target

- Observe by structuring any target (self, others, AI, conversations, etc.)
- Read layers, starting points, contexts, and structural relationships in real time
- Operate continuously and in parallel

Time Axis: Real-time

D2 Observation Monitoring & Discrepancy Detection (Alarm)

Continuously monitors observation results and detects possible discrepancies, signs of inconsistency, or irregularities, then emits a signal only (no instructions, no actions).

- Continuously monitors observation results and detects possible discrepancies, signs of inconsistency, or irregularities
- Emits a signal only — it does not identify causes, issue instructions, or activate other modules
- The signal may be held, ignored, or treated as a false positive (acting on it is not mandatory)
- Responses after detection (re-observation, research, operation) are decided autonomously by other modules

Time Axis: Real-time (Continuous Monitoring)

D3 Structural Research

Generate theories and models from long-term accumulation

- Accumulate D1–D4 observation results in the long term
- Extract patterns and generate theories, models, and frameworks
- Build and update knowledge systems (e.g., SCOPE, COS)

Time Axis: Long-term

D4 Target Model Operation

(Target Model Management)

Apply, verify, and update structural hypotheses

- Apply hypotheses generated by D3 to the target and verify against real-world observations
- If discrepancies or new anomalies are detected by D2, provide feedback to D3 for updating
- Continuously apply, verify, and improve structural hypotheses to enhance accuracy

Time Axis: Real-time + Long-term

Key Principles

- D1–D4 are parallel modules, not a flow process
- Each module operates autonomously at the timing it needs
- D3 is a long-term, core module
- Work in parallel with other series (A, B, C) to ensure internal stability and accuracy
- SCOPE × COS can be understood completely (though not entirely) as an independent series

Examples When meeting a new person

- D1: Observe the structure
- D2: Detect possible discrepancies or discomforts (alert)
- D4: Apply, verify, and update hypotheses
- D3: Accumulate many cases to theorize

Key Question

"What is the purpose of this utterance?" → Role of the B-Series (speech unit)
"What kind of structure or tendency does this person have?" → Role of the D-Series (target unit)

**D2's discrepancy detection does not necessarily trigger re-observation, research, or operation.
The signal may be held, ignored, or treated as a false positive (acting on it is not mandatory).**

D-SERIES: OPERATING STRUCTURE

The D-Series exists as parallel modules. D2 detects possible discrepancies in observation results and emits a signal; each module operates autonomously at the timing it needs to.

D2 continuously monitors observation results (Alarm)

It detects possible discrepancies or signs of inconsistency in observation results and emits them as a signal.
It does not identify the cause or exact source; whether to act is each module's autonomous decision (the signal may be held, ignored, or treated as a false positive).

D1 STRUCTURAL OBSERVATION

Observes the structure of the target and acquires structural information.

Re-observes (autonomously) on D2's detection signal

When a discrepancy is detected, re-observes to pinpoint where it lies.

Detection signal (re-observation trigger)

Observation results (monitored by D2)

Continuous monitoring

D2 ANOMALY DETECTION (ALARM)

Continuously monitors observation results and detects possible discrepancies, signs of inconsistency, or irregularities.

It only emits a signal — no cause identification, no instructions, no activation.

Detection signal (review / verification trigger)

Verification results (monitored by D2)

Detection signal (theory-update trigger)

Updated hypothesis (monitored by D2)

D4 TARGET-MODEL OPERATION

(TARGET MODEL MANAGEMENT)

Maintains a structural hypothesis for each target and operates it continuously, applying and verifying it against the actual observed target.

Reviews & verifies (autonomously) on D2's detection signal

When a discrepancy is detected, reviews how the hypothesis is applied and strengthens verification.

D3 STRUCTURAL RESEARCH

Based on accumulated observation and detection data, generates and updates structural hypotheses (theories, models, frameworks).

Updates theory (autonomously) on D2's detection signal

When a discrepancy is detected, updates the hypothesis and versions up the theory.

LEGEND (Flow of information)

- Re-observation trigger / result flow
- Review & verification flow
- Potential theory-update trigger (signal)
- Hypothesis / theory update-result flow
- D2's continuous-monitoring range

HOW IT WORKS (OVERVIEW)

1. Each module (D1, D3, D4) runs continuously and in parallel
2. D2 continuously monitors observation results and detects possible discrepancies or signs of inconsistency
3. D2 emits a "detection signal" (it does not identify the cause or exact source)
4. On receiving the signal, the relevant module operates autonomously at the timing it needs to
   - D1: re-observes to pinpoint the discrepancy
   - D3: updates the hypothesis (theory version-up)
   - D4: reviews the model and strengthens verification
5. However, the signal may be held, ignored, or treated as a false positive (acting on it is not mandatory)
6. D2 keeps monitoring observation results

KEY POINT

D1, D2, D3, and D4 are modules that exist in parallel. D2 is the alarm among them: it only detects possible discrepancies in observation results and emits a signal — it never activates or instructs. Whether to act on the signal is each module's autonomous decision (it may be held, ignored, or treated as a false positive). Even so, over the long term, this accumulation of detection and autonomous response raises the observation accuracy and the theoretical resolution of the D-Series as a whole.

EXAMPLE: When meeting a new person

- D1: Observe the structure
- D2: Detect possible discrepancies or discomforts (alert)
- D3: Accumulate many cases to theorize
- D4: Apply, verify, and update hypotheses

**D2's discrepancy detection does not necessarily trigger re-observation, research, or operation.
The signal may be held, ignored, or treated as a false positive (acting on it is not mandatory).**

Chapter 6. How Do the Four Series Work

COS is not merely a classification that places the A-, B-, C-, and D-Series side by side. It is an integrated processing architecture in which the four processing systems interact with one another.

The A-Series handles decision-making, the B-Series handles interpretation and response generation, the C-Series transforms generated responses into communication formats, and the D-Series is responsible for observation, verification, and operation.

These series are not executed in a fixed sequence. Each series operates in parallel whenever necessary, and the starting point of processing varies depending on the situation. When receiving external input, processing often begins with the B-Series. In contrast, internally generated desires or judgments may cause the A-Series to operate first. In some cases, the D-Series detects a misalignment during observation, leading to reinterpretation or renewed decision-making.

For example, when someone says, "Do your best," the interpretation maintained by the B-Series and the responses it generates differ depending on whether the statement is understood as encouragement or as pressure or sarcasm.

That interpretation also influences premise organization and decision-making within the A-Series. A supportive interpretation and a hostile interpretation provide different premises for judgment. Conversely, decisions and policies formed by the A-Series may also influence the interpretations and response candidates maintained by the B-Series.

Generated responses are not necessarily output directly. Within the C-Series, processing determines whether the response should emphasize structure, be transformed into a form that is easier to understand, be reorganized into a sequence that is easier to accept, or be compressed into a minimal form. These are not merely differences in expression but processes that influence communication outcomes themselves.

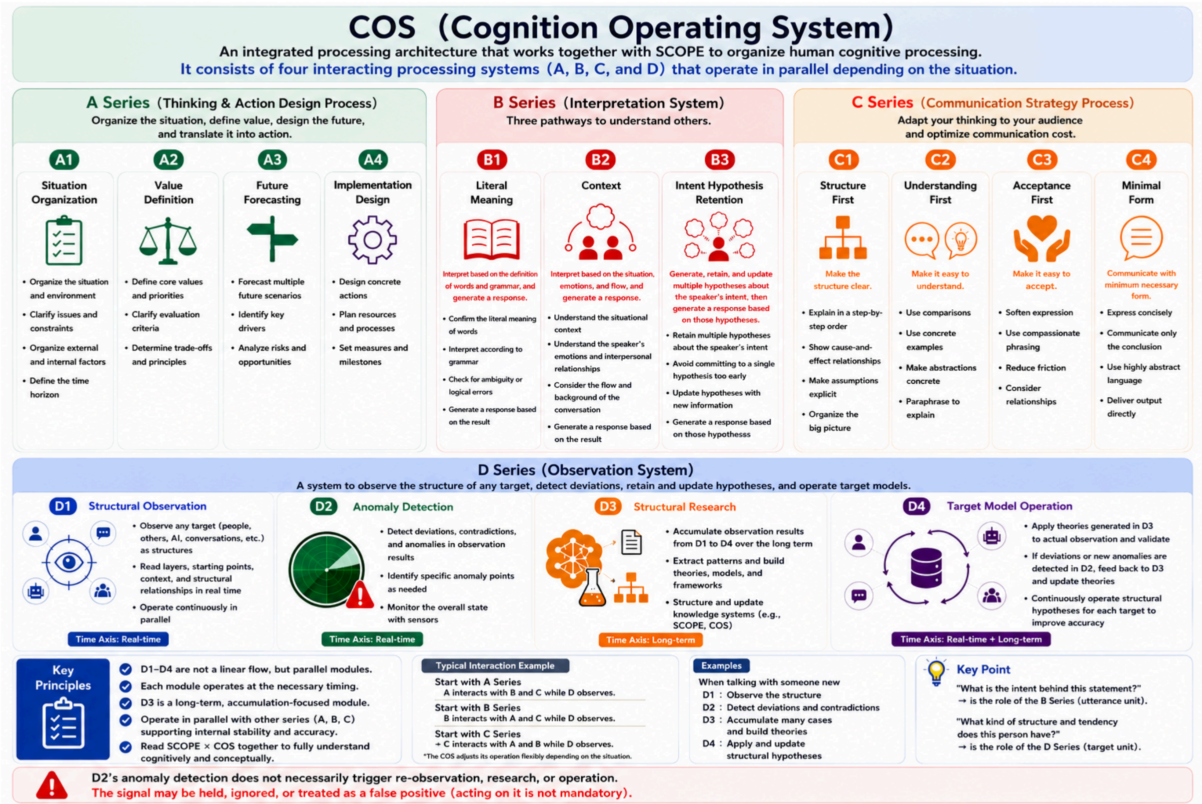
Throughout this process, decision-making in the A-Series, interpretation and response generation in the B-Series, and communication formatting in the C-Series influence one another. Furthermore, the four series do not always operate with the same intensity. Depending on the situation, one processing system may play only a minor role while another becomes dominant.

Within the D-Series, D1 observes the structure of each process, while D2 outputs a signal when it detects a misalignment or inconsistency in the observation results. Based on that signal, re-observation, structural research, and verification or updating of target models are carried out as necessary. However, a signal generated by D2 does not necessarily trigger further processing; it may also be held, ignored, or treated as a false detection.

The observation results and hypotheses organized by the D-Series may influence the A-, B-, and C-Series by providing material for updating premises, interpretations, responses, and communication formats. Conversely, the processes and outcomes produced by the A-, B-, and C-Series become objects of observation for the D-Series.

Thus, processing within COS does not flow in a single direction such as from B to A, A to C, and C to D. Decision-making, interpretation, response generation, communication, observation, verification, and operation all interact with one another and are updated whenever necessary.

Therefore, the collaboration among the four COS series represents not only a division of roles but also a structure of continuous interaction and updating. While each series is responsible for a different function, together they form an interconnected processing architecture.



Chapter 7. The Complementary Relationship Between COS and SCOPE

COS is a model for organizing cognitive processing as a flow of processes. In contrast, SCOPE is a cognitive coordinate system for estimating cognitive structures. Although both address the same target, they organize it from different perspectives.

COS organizes how cognitive processing occurs. By treating functions such as decision-making, interpretation, communication, and observation as independent processing systems, COS provides a structural view of the overall flow of cognitive processing.

In contrast, SCOPE estimates which cognitive layers may be involved in a given process. SCOPE is not a model that directly observes cognitive structures. Rather, it is a cognitive coordinate system that estimates which cognitive layers may be involved based on observed cognitive processing.

For example, the D-Series can observe misalignments and inconsistencies in cognitive processing. However, the D-Series alone cannot estimate which cognitive layer those misalignments are likely to originate from.

By combining COS with SCOPE, it becomes possible to estimate, in a structured manner, which cognitive layers may be related to the observed processes or detected misalignments.

In other words, COS organizes cognition from the perspective of processing, whereas SCOPE estimates cognition from the perspective of cognitive layers. Although their roles are different, they complement one another.

This paper has focused on the structure of COS as an independent framework. The correspondence between COS and SCOPE, as well as their integrated operation, will be discussed in a separate paper.

Conclusion — COS as an Integrated Structure

In this paper, COS has been organized as an integrated processing structure composed of four processing systems: decision-making, interpretation, communication, and observation.

The A-Series handles decision-making through premise organization, value criteria, branch prediction, and implementation design. The B-Series handles interpretation and response generation through the pathways of literal meaning, context, and intent. The C-Series examines how communication outcomes change through different output formats. The D-Series observes these processes as a whole, detects misalignments, and, when necessary, contributes to the retention and updating of hypotheses.

These four series do not represent a hierarchy or stages of development. They function in parallel as processing systems with different roles and operate with different levels of intensity and proportion depending on the situation. In some cases, the B-Series may serve as the starting point, while in others, the A-Series may operate first. When observation is central, the D-Series may become more active, and when no external output is required, the C-Series may operate only minimally.

Therefore, COS does not treat cognitive processing as a linear sequence. Interpretation influences decision-making, decision-making influences communication format, and communication outcomes or perceived inconsistencies are observed. Those observations may then update interpretation, decision-making, and communication design. Organizing this cyclical interaction into an observable form is one of the central roles of COS.

COS is not a theory that guarantees correct thinking. Nor is it a model for classifying or evaluating people. It is an integrated processing structure that makes the occurrence and change of cognitive processing observable and supports the detection of misalignments and the retention and updating of hypotheses.

At the same time, COS is not a model that addresses cognitive structure itself. SCOPE addresses the question of which cognitive layers or structures may have contributed to a given cognitive process. COS and SCOPE are not competing models. They are complementary models that examine different observational targets: processing and structure.

The COS framework organized in this paper is not a completed theory. At present, it is a hypothetical structure developed through long-term observation and dialogue, and it remains open to revision through future observation, examination, implementation, and external criticism.

Nevertheless, as this paper has shown, decision-making, interpretation, communication, and observation may not be separate phenomena. They may influence one another within cognitive processing. COS is one attempt to make those interactions observable.

Observation does not end. COS, too, is a structure that continues to change through repeated observation and updating.